ARTIFICIAL INTELLIGENCE OVERVIEW

Artificial Intelligence (AI) refers to the simulation of human intelligence in machines.
The goal of AI is to develop systems capable of performing tasks that typically require human cognition, such as visual perception, speech recognition, decision-making, and language translation.

Machine Learning is a subset of AI that focuses on building applications that learn from data and improve their accuracy over time without being explicitly programmed to do so.

SUPERVISED AND UNSUPERVISED LEARNING

Supervised learning algorithms are trained using labeled datasets, where the desired output is known.
Common algorithms include Linear Regression, Support Vector Machines, and Decision Trees.
Supervised learning is widely used for classification and regression tasks like spam detection and price pre

Unsupervised learning uses unlabeled data to discover hidden patterns and clusters without human guidan
Examples include K-Means Clustering and Principal Component Analysis (PCA).

DEEP LEARNING AND NEURAL NETWORKS

Deep Learning is a specialized subfield of Machine Learning based on Artificial Neural Networks with repre
The term deep refers to the number of layers in the network.
A typical deep neural network consists of an input layer, multiple hidden layers, and an output layer.

Convolutional Neural Networks (CNNs) excel at image processing, while Recurrent Neural Networks (RNN
Transformers are designed for sequential data and natural language processing.

RETRIEVAL-AUGMENTED GENERATION (RAG)

Retrieval-Augmented Generation (RAG) is an architectural framework that improves the accuracy of Large
Instead of relying solely on training weights, RAG retrieves relevant facts from an external knowledge base
or private document collection before generating a response.

Key advantages of RAG include:
1. Reduction in hallucination by grounding responses in verified source texts.
2. Access to domain-specific and up-to-date data without expensive fine-tuning.
3. Verifiable source citations and page transparency for users.